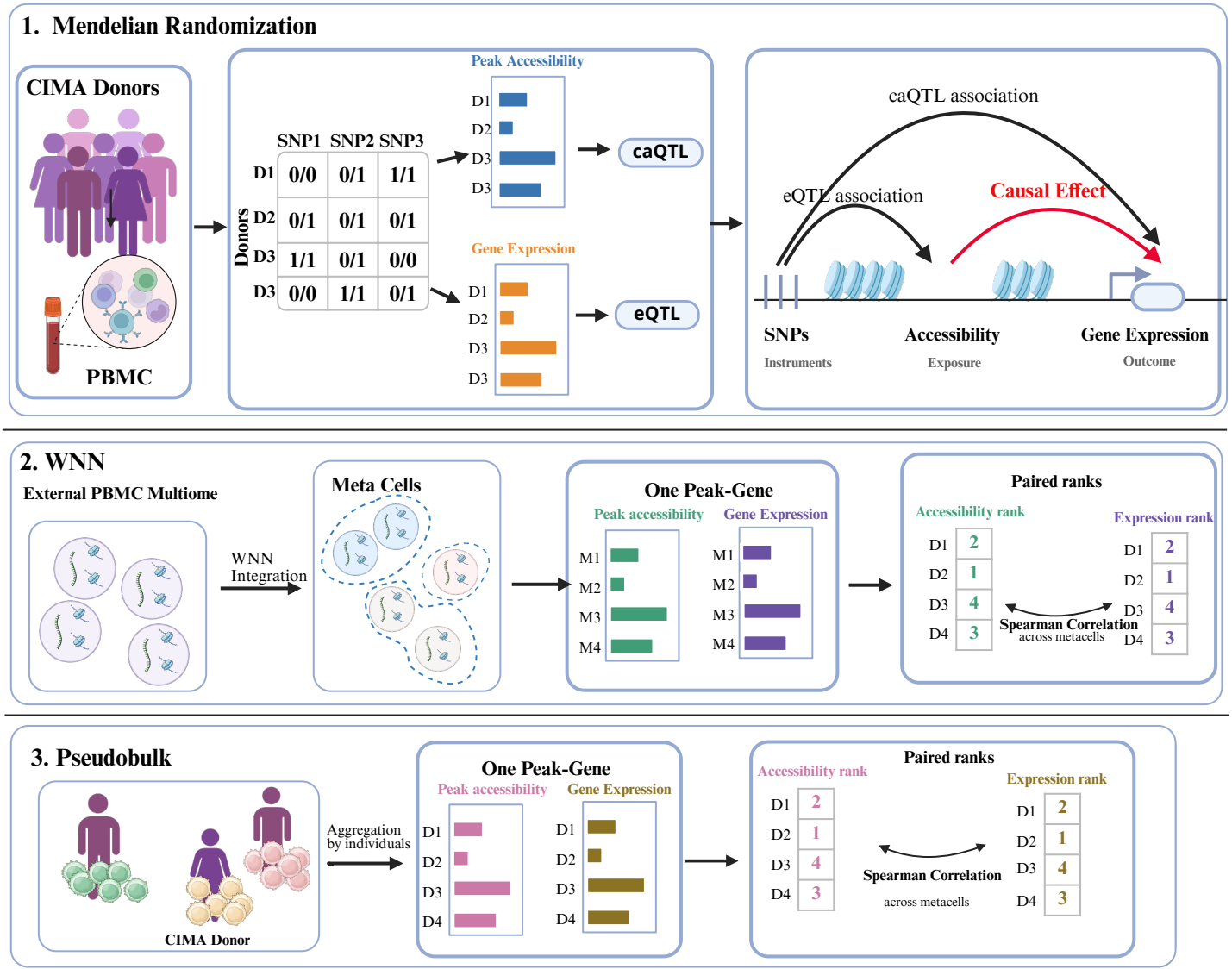

1. Mendelian Randomization
CIMA Donors
PBMC
Donors
SNP1 SNP2 SNP3
D1 0/0 0/1 1/1
D2 0/1 0/1 0/1
D3 1/1 0/1 0/0
D3 0/0 1/1 0/1
Peak Accessibility
D1
D2
D3
D3
caQTL
Gene Expression
D1
D2
D3
D3
eQTL
caQTL association
eQTL association
Causal Effect
SNPs
Instruments
Accessibility
Exposure
Gene Expression
Outcome
2. WNN
External PBMC Multiome
WNN Integration
Meta Cells
One Peak-Gene
Peak accessibility
Gene Expression
M1
M2
M3
M4
Paired ranks
Accessibility rank
Expression rank
D1 2
D2 1
D3 4
D4 3
Spearman Correlation
across metacells
3. Pseudobulk
CIMA Donor
Aggregation by individuals
One Peak-Gene
Peak accessibility
Gene Expression
D1
D2
D3
D4
Paired ranks
Accessibility rank
Expression rank
D1 2
D2 1
D3 4
D4 3
Spearman Correlation
across metacells